1 Effects of Gravitation

Some guide the course of wand'ring orbs on high,
Or roll the planets through the boundless sky.
Some less refined, beneath the moon's pale light
Pursue the stars that shoot athwart the night,
Or suck the mists in grosser air below,
Or dip their pinions in the painted bow
Or blow fierce tempests on the wintry main,
Or o'er the glebe distill the kindly rain.

Alexander Pope, *The Rape of the Lock*

We now return to physics, and apply what we have learned in the last chapter to determine the effects of gravitation on the equations of mechanics and electrodynamics. The technique to be used is that afforded by the Principle of General Covariance: We must first write the equations as they hold in special relativity, then decide how each quantity in the equations is to transform under general coordinate transformations, and then replace $\eta_{\mu\nu}$ with $g_{\mu\nu}$ and all derivatives with covariant derivatives. The resulting equations will be generally covariant and true in the absence of gravitation, and hence true in arbitrary gravitational fields, provided that the system in question is small enough compared with the scale of the fields.

1.1 Particle Mechanics

A particle not under the influence of any force will in special relativity have constant four-velocity u^a and constant spin S_a, that is,

$$\frac{\mathrm{d}u^a}{\mathrm{d}\tau} = 0, \qquad \left(u^a \equiv \frac{\mathrm{d}\xi^a}{\mathrm{d}\tau}\right), \tag{5.1.1}$$

$$\frac{\mathrm{d}S_a}{\mathrm{d}\tau} = 0. \tag{5.1.2}$$

Recall that S_a is defined in the rest frame of the particle to have the time-first components $\{0, \mathbf{S}\}$, so that in an arbitrary Lorentz frame it satisfies the further relation

$$S_a u^a = 0. \tag{5.1.3}$$

In order to make these equations generally covariant, we define vectors u^μ and S_μ in a general coordinate system x^μ by

$$u^\mu \equiv \frac{\partial x^\mu}{\partial \xi^a} u^a = \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau}, \tag{5.1.4}$$

$$S_\mu = \frac{\partial \xi^a}{\partial x^\mu} S_a, \tag{5.1.5}$$

where u^a and S_a are components of $\mathbf{u}$ and $\mathbf{S}$ in the freely falling coordinate system ξ^a. Although u^μ and S_μ are vectors, $\mathrm{d}u^\mu/\mathrm{d}\tau$ and $\mathrm{d}S_\mu/\mathrm{d}\tau$ are not, but we saw in Section **??**

that there can be defined vector derivatives $\frac{Du^\mu}{D\tau}$ and $\frac{DS_\mu}{D\tau}$, which reduce when $\Gamma^\lambda{}_{\mu\nu} = 0$ to the ordinary derivatives $\mathrm{d}u^\mu / \mathrm{d}\tau$ and $\mathrm{d}S_\mu / \mathrm{d}\tau$. The correct equations for the particle position and spin are dictated by the Principle of General Covariance to be

$$\frac{Du^\mu}{D\tau} = 0, \qquad \frac{DS_\mu}{D\tau} = 0, \tag{5.1.6}$$

or, in more detail,

$$\frac{\mathrm{d}u^\mu}{\mathrm{d}\tau} + \Gamma^\mu{}_{\nu\lambda} u^\nu u^\lambda = 0, \tag{5.1.7}$$

$$\frac{\mathrm{d}S_\mu}{\mathrm{d}\tau} - \Gamma^\lambda{}_{\mu\nu} u^\nu S_\lambda = 0. \tag{5.1.8}$$

In addition, Eq. (5.1.3) becomes now

$$S_\mu u^\mu = 0. \tag{5.1.9}$$

To repeat the reasoning of Section **??**, these equations are true in the presence of gravitational fields because they are generally covariant and true in the absence of gravitation, reducing when $\Gamma^\lambda{}_{\mu\nu}$ vanishes to Eqs. (5.1.1)–(5.1.3). That is, the Principle of Equivalence tells us that there are locally inertial coordinate systems in which Eqs. (5.1.6)–(5.1.9) are valid (provided always that our particle is sufficiently small), and general covariance then ensures that these equations hold in the laboratory reference frame.

We recognize in Eqs. (5.1.7) and (5.1.8) the differential equations for parallel transport of the vectors u^μ and S_μ. Since $u^\mu = \mathrm{d}x^\mu / \mathrm{d}\tau$, Eq. (5.1.7) is nothing but the familiar equation for free fall, derived previously by differentiating Eq. (5.1.4) with respect to τ and using Eq. (5.1.1); it should be evident that a good deal of work is saved by using general covariance instead of our previous direct approach. Equation (5.1.8) describes the precession of gyroscopes in free fall, and is further discussed in Chapter 9. For the present, we note that $S_\mu S^\mu$ is constant, because the ordinary derivative of a scalar is the same as its covariant derivative:

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\left(S_\mu S^\mu\right) = \frac{D}{D\tau}\left(S_\mu S^\mu\right) = 0. \tag{5.1.10}$$

If the particle is not in free fall, then $\frac{Du^\mu}{D\tau}$ does not vanish, and instead of Eq. (5.1.7) we have

$$\frac{Du^\mu}{D\tau} \equiv \frac{f^\mu}{m}, \tag{5.1.11}$$

where m is the particle mass and f^μ is a contravariant force vector. This can also be written as

$$m\frac{\mathrm{d}^2 x^\mu}{\mathrm{d}\tau^2} = f^\mu - m\Gamma^\mu{}_{\nu\lambda}\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau}\frac{\mathrm{d}x^\lambda}{\mathrm{d}\tau}.$$

The term containing $m\Gamma^\mu{}_{\nu\lambda}$ evidently plays the role of a gravitational force. We can always calculate f^μ if we know its value f^a_{f} in freely falling frames ξ^a, for the requirement that f^μ behave as a vector gives it uniquely as

$$f^\mu = \frac{\partial x^\mu}{\partial \xi^a} f^a_{\mathrm{f}}. \tag{5.1.12}$$

The electromagnetic force will be determined in the next section.

It sometimes happens that a particle is acted on by a force f^μ without experiencing any torque. In this event, an observer in a locally inertial coordinate frame that is momentarily at rest with respect to the particle will see no precession of the spin axis; that is, $\mathrm{d}\mathbf{S}/\,\mathrm{d}t$ will vanish. But in this particular coordinate system $\mathrm{d}\mathbf{x}/\,\mathrm{d}t$ also vanishes, so we may write the condition of zero torque as a Lorentz-invariant statement

$$\frac{\mathrm{d}S^a}{\mathrm{d}\tau} \propto u^a,$$

and this will be valid in any locally inertial coordinate system, comoving or not. Now, what is the constant of proportionality? Let us set

$$\frac{\mathrm{d}S^a}{\mathrm{d}\tau} = \Phi u^a.$$

We recall that S_a is defined so that

$$S_a u^a = 0,$$

and therefore

$$0 = \frac{\mathrm{d}}{\mathrm{d}\tau}\left(S_a u^a\right) = \Phi u_a u^a + S_a \frac{\mathrm{d}u^a}{\mathrm{d}\tau},$$

so

$$\Phi = S_a \frac{\mathrm{d}u^a}{\mathrm{d}\tau} = S_a \frac{f^a}{m}.$$

The spin vector therefore suffers a change given by

$$\frac{\mathrm{d}S^a}{\mathrm{d}\tau} = \left(S_b \frac{f^b}{m}\right) u^a. \tag{5.1.13}$$

This phenomenon is known as the *Thomas precession*.[1] If we now turn on a gravitational field, Eq. (5.1.13) and the Principle of General Covariance tell us that the spin precesses according to the rule

$$\frac{DS^\mu}{D\tau} = \left(S_\nu \frac{f^\nu}{m}\right) u^\mu = S_\nu \frac{Du^\nu}{D\tau} u^\mu. \tag{5.1.14}$$

A vector obeying this differential equation is said to be defined by *Fermi transport*;[2] parallel transport is the special case for $f^\mu = 0$.

1.2 Electrodynamics

We recall that in the absence of gravitational fields the Maxwell equations of electrodynamics can be written as

$$\partial_a F^{ab} = -J^b, \tag{5.2.1}$$

$$\partial_{[a} F_{bc]} = 0, \tag{5.2.2}$$

where $J^a = (\rho, \mathbf{J})$ is the current four-vector and F^{ab} is the field-strength tensor, with $F^{0i} = E_i$, $F^{ij} = \varepsilon^{ijk} B_k$, and so on. (See Section **??**.) Suppose that we define $F^{\mu\nu}$ and J^μ in general coordinates by the requirements that they reduce to F^{ab} and J^a in locally inertial Minkowskian coordinates, and that they behave as tensors under general coordinate transformations. That is, if F^{ab} and J^a are the values measured in a locally inertial frame, then

$$F^{\mu\nu} \equiv \frac{\partial x^\mu}{\partial \xi^a}\frac{\partial x^\nu}{\partial \xi^b}F^{ab}, \qquad J^\mu \equiv \frac{\partial x^\mu}{\partial \xi^a}J^a.$$

We can then make Eqs. (5.2.1) and (5.2.2) generally covariant by replacing all derivatives by covariant derivatives:

$$\nabla_\mu F^{\mu\nu} = -J^\nu, \tag{5.2.3}$$

$$\nabla_{[\mu}F_{\nu\rho]} = 0, \tag{5.2.4}$$

it being now understood that indices are to be raised and lowered with $g_{\mu\nu}$ instead of $\eta_{\mu\nu}$; that is,

$$F_{\mu\nu} \equiv g_{\mu\rho}g_{\nu\sigma}F^{\rho\sigma}. \tag{5.2.5}$$

Since $F^{\mu\nu}$ and $F_{\mu\nu}$ are antisymmetric, we may use Eqs. (**??**) and (**??**) to rewrite the Maxwell equations as

$$\partial_\mu\big(\sqrt{-g}\,F^{\mu\nu}\big) = -\sqrt{-g}\,J^\nu, \tag{5.2.6}$$

$$\partial_{[\mu}F_{\nu\rho]} = 0. \tag{5.2.7}$$

Equations (5.2.3)–(5.2.7) are true in the absence of gravitation and generally covariant and hence, according to the Principle of General Covariance, also true in arbitrary gravitational fields.

The electromagnetic force on a particle of charge e is given in the absence of gravitation by Eq. (**??**):

$$f^a = eF^a{}_b\frac{\mathrm{d}\xi^b}{\mathrm{d}\tau}. \tag{5.2.8}$$

We immediately conclude that in general coordinates the electromagnetic force in an arbitrary gravitational field is

$$f^\mu = eF^\mu{}_\nu\frac{\mathrm{d}x^\nu}{\mathrm{d}\tau}, \tag{5.2.9}$$

where of course

$$F^\mu{}_\nu \equiv g_{\nu\lambda}F^{\mu\lambda}. \tag{5.2.10}$$

Once again, we are using the Principle of General Covariance; Eq. (5.2.9) obviously reduces to Eq. (5.2.8) in locally inertial Minkowskian coordinates, and it is generally covariant, because f^μ is a vector (Section 1.1), $\mathrm{d}x^\nu/\,\mathrm{d}\tau$ is a vector, and $F^\mu{}_\nu$ is defined as a tensor; therefore Eq. (5.2.9) is true.

It is instructive to evaluate the current vector J^μ. In special relativity it is

$$J^a = \sum_n e_n\int \delta^4(\xi - \xi_n)\,\mathrm{d}\xi_n^a, \tag{5.2.11}$$

the integral being taken along the trajectory of the nth particle. [See Eq. (**??**).] The four-dimensional delta function in a general coordinate system is defined by

$$\int \mathrm{d}^4x\,\Phi(x)\delta^4(x-y) = \Phi(y). \tag{5.2.12}$$

Since $\sqrt{-g}\,\mathrm{d}^4x$ is a scalar, $(-g)^{-1/2}\delta^4(x-y)$ must be a scalar, which of course reduces to the ordinary delta function in special relativity, where $-g=1$. (In some works it is this scalar that is defined as the delta function.) Thus the contravariant vector that reduces to J^a in the absence of gravitation is

$$J^\mu(x) = \frac{1}{\sqrt{-g(x)}}\sum_n e_n \int \delta^4(x-x_n)\,\mathrm{d}x_n^\mu. \tag{5.2.13}$$

Note that the conservation law $\partial_a J^a = 0$ of special relativity becomes in general relativity $\nabla_\mu J^\mu = 0$, or, using Eq. (**??**),

$$\partial_\mu\big(\sqrt{-g}\,J^\mu\big) = 0. \tag{5.2.14}$$

The factor $(-g)^{-1/2}$ in Eq. (5.2.13) is just what is needed to cancel the $\sqrt{-g}$ in Eq. (5.2.14), so that Eq. (5.2.14) still just expresses the constancy of the e_n.

1.3 Energy-Momentum Tensor

The density and current of energy and momentum were united in Section **??** into a symmetric tensor $T^{\mu\nu}$ satisfying the conservation equations

$$\partial_\mu T^{\mu\nu} = G^\nu, \tag{5.3.1}$$

where G^ν is the density of the external force f^ν acting on the system. (For an isolated system, $G^\nu = 0$.) Define $T^{\mu\nu}$ and G^ν as contravariant tensors that reduce to the special-relativistic $T^{\mu\nu}$ and G^ν in the absence of gravitation. Then the generally covariant equation that agrees with Eq. (5.3.1) in locally inertial systems is

$$\nabla_\mu T^{\mu\nu} = G^\nu, \tag{5.3.2}$$

or, using Eq. (**??**),

$$\frac{1}{\sqrt{-g}}\,\partial_\mu\big(\sqrt{-g}\,T^{\mu\nu}\big) = G^\nu - \Gamma^\nu{}_{\mu\lambda}T^{\mu\lambda}. \tag{5.3.3}$$

The factor $\sqrt{-g}$ is familiar from electrodynamics, and arises from the fact that the invariant volume is $\sqrt{-g}\,\mathrm{d}^4x$. In contrast, the second term on the right represents a gravitational force density. Just as we would expect, this force depends on the system on which it acts only through the energy-momentum tensor.

For a system of point particles the special-relativistic energy-momentum tensor is given in Section **??** as

$$T^{\mu\nu}(x) = \sum_n m_n \int \frac{\mathrm{d}x_n^\mu}{\mathrm{d}\tau}\,\mathrm{d}x_n^\nu\,\delta^4(x-x_n)\,, \tag{5.3.4}$$

the integral again being taken along the particle trajectory. Following precisely the same reasoning as we did for J^μ in the last section, we conclude that the contravariant tensor that agrees with Eq. (5.3.4) in the absence of gravitation is

$$T^{\mu\nu}(x) = \frac{1}{\sqrt{-g(x)}}\sum_n m_n \int \frac{\mathrm{d}x_n^\mu}{\mathrm{d}\tau}\,\mathrm{d}x_n^\nu\,\delta^4(x-x_n)\,. \tag{5.3.5}$$

For an electromagnetic field $F^{\mu\nu}$ the special-relativistic energy-momentum tensor was calculated in Section **??** as

$$T^{\mu\nu}_{\text{em}} = F^{\mu}{}_{\rho} F^{\nu\rho} - \frac{1}{4}\eta^{\mu\nu} F_{\rho\sigma} F^{\rho\sigma}. \tag{5.3.6}$$

It takes no effort to see that the contravariant tensor that agrees with Eq. (5.3.6) in the absence of gravitation is

$$T^{\mu\nu}_{\text{em}} = F^{\mu}{}_{\rho} F^{\nu\rho} - \frac{1}{4}g^{\mu\nu} F_{\rho\sigma} F^{\rho\sigma}. \tag{5.3.7}$$

For a system consisting of particles and radiation, the energy-momentum tensor is the sum of Eqs. (5.3.5) and (5.3.7).

Returning for a moment to the purely material energy-momentum tensor (5.3.5), we easily compute that

$$\int T^{\mu 0}\sqrt{-g}\ \mathrm{d}^3x = \sum_n m_n \frac{\mathrm{d}x_n^\mu}{\mathrm{d}\tau},$$

the sum running over all particles in the volume of integration. This suggests that $T^{\mu 0}\sqrt{-g}$ is to be regarded in general as the spatial density of energy and momentum. In particular, we are tempted to define the energy, momentum, and angular momentum for an arbitrary system by

$$P^\mu \equiv \int T^{\mu 0}\sqrt{-g}\ \mathrm{d}^3x, \tag{5.3.8}$$

$$J^{\mu\nu} \equiv \int \left(x^\mu T^{\nu 0} - x^\nu T^{\mu 0}\right)\sqrt{-g}\ \mathrm{d}^3x. \tag{5.3.9}$$

However, these quantities are *not* contravariant tensors and are *not* conserved, because $T^{\mu\nu}\sqrt{-g}$ is not conserved; that is, $\partial_\nu(\sqrt{-g}\,T^{\mu\nu})$ does not vanish, owing to the exchange of energy and momentum between matter and gravitation.

1.4 Hydrodynamics and Hydrostatics

In the absence of gravitation, the energy-momentum tensor of a perfect fluid is that given by Eq. (**??**):

$$T^{\mu\nu} = p\eta^{\mu\nu} + (\rho + p)u^\mu u^\nu, \tag{5.4.1}$$

where u^μ is the fluid four-velocity,

$$u^0 = (1 - \mathbf{v}^2)^{-1/2}, \qquad \mathbf{u} = \mathbf{v}u^0.$$

The contravariant tensor that reduces to Eq. (5.4.1) in the absence of gravitation is

$$T^{\mu\nu} = pg^{\mu\nu} + (\rho + p)u^\mu u^\nu, \tag{5.4.2}$$

where u^μ is the local value of $\mathrm{d}x^\mu/\mathrm{d}\tau$ for a comoving fluid element. Note that p and ρ are always defined as the pressure and energy density measured by an observer in a locally inertial frame that happens to be moving with the fluid at the instant of measurement, and are therefore scalars. The conditions of energy-momentum conservation give the hydrodynamic equations

$$0 = \nabla_\nu T^{\mu\nu} = g^{\mu\nu}\partial_\nu p + \frac{1}{\sqrt{-g}}\,\partial_\nu\left[\sqrt{-g}\,(\rho+p)u^\mu u^\nu\right] \\ + \Gamma^\mu{}_{\nu\lambda}(\rho+p)u^\nu u^\lambda. \tag{5.4.3}$$

The last term represents the gravitational force on the system. Note also that since $\eta_{\mu\nu}u^\mu u^\nu = -1$ in the absence of gravitation, we must in the presence of gravitation have

$$g_{\mu\nu}u^\mu u^\nu = -1. \tag{5.4.4}$$

Consider as an example the case of a fluid in hydrostatic equilibrium. Since it is not moving, Eq. (5.4.4) gives

$$u^0 = (-g_{00})^{-1/2}, \qquad u^i = 0.$$

Furthermore, all temporal derivatives of $g_{\mu\nu}$, p, or ρ vanish. In particular,

$$\Gamma^\nu{}_{00} = -\frac{1}{2}g^{\nu\lambda}\partial_\lambda g_{00},$$

and

$$\partial_\nu\left[(\rho+p)u^\mu u^\nu\right] = 0.$$

Multiplying Eq. (5.4.3) by $g_{\mu\lambda}$ gives then

$$-\partial_\lambda p = (\rho+p)\partial_\lambda \ln\sqrt{-g_{00}}. \tag{5.4.5}$$

This is trivial for $\lambda = 0$, whereas for λ spacelike it is nothing but the ordinary nonrelativistic equation of hydrostatic equilibrium, except that $\rho + p$ appears instead of the mass density and $\ln\sqrt{-g_{00}}$ appears instead of the gravitational potential. This equation is soluble if p is given as a function of ρ. We then find that

$$\int \frac{\mathrm{d}p(\rho)}{p(\rho)+\rho} = -\ln\sqrt{-g_{00}} + \text{constant}. \tag{5.4.6}$$

For instance, if $p(\rho)$ is given by a power law

$$p(\rho) \propto \rho^N, \tag{5.4.7}$$

then Eq. (5.4.6) gives for $N \neq 1$

$$\frac{\rho+p}{\rho} \propto (-g_{00})^{(1-N)/(2N)}, \tag{5.4.8}$$

and for $N = 1$

$$\rho \propto (-g_{00})^{-(p+\rho)/(2p)}. \tag{5.4.9}$$

This, incidentally, shows that gravitation can never produce hydrostatic equilibrium in a finite highly relativistic fluid with $p = \rho/3$, for then Eq. (5.4.9) gives

$$\rho \propto (-g_{00})^{-2}. \tag{5.4.10}$$

Since ρ must vanish outside the fluid, g_{00} would have to become singular at its surface.

References

1. L. H. Thomas, *Nature* **117**, 514 (1926).
2. E. Fermi, *Atti. R. Accad. Rend. Cl. Sc. Fis. Mat. Nat.* **31**, 21 (1922).